

CHARLEVOIX, MI, USA

WMO: 727434

Lat: 45.305N

Lon: 85.275W

Elev: 646

StdP: 14.36

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 04870

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.2	4.9	-7.4	3.8	1.6	-2.2	5.0	6.9	28.9	20.8	26.3	20.8	6.2	100	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	84.3	70.3	81.9	69.1	79.4	67.8	73.2	80.8	71.6	78.7	70.2	76.7	9.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.0	117.0	77.1	69.7	111.7	75.8	67.9	105.0	74.6	37.2	80.9	35.8	78.6	34.5	76.0	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.9	21.2	18.9	DB	-8.3	90.8	7.2	3.1	-13.4	93.1	-17.6	94.9	-21.7	96.6	-26.9	98.8	(4)
(5)			WB	-8.9	76.0	7.0	1.7	-13.9	77.2	-18.0	78.2	-21.9	79.1	-26.9	80.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.5	22.4	21.9	30.1	41.2	52.6	62.5	68.4	67.9	61.9	49.2	38.1	28.8	(6)	
	DBStd	18.51	9.23	9.58	11.12	8.73	8.41	7.14	5.62	5.12	7.29	8.06	7.74	7.72	(7)	
	HDD50	3768	857	787	626	291	64	2	0	0	3	115	366	657	(8)	
	HDD65	7479	1322	1207	1084	715	397	129	29	29	144	493	808	1122	(9)	
	CDD50	2143	0	0	8	27	147	376	572	554	359	92	8	0	(10)	
	CDD65	376	0	0	0	0	14	54	136	117	51	4	0	0	(11)	
	CDH74	2409	0	0	6	11	121	420	913	620	290	28	0	0	(12)	
	CDH80	476	0	0	0	1	25	100	191	102	55	2	0	0	(13)	
(14) Wind	WSAvg	8.6	10.6	9.8	8.3	8.7	7.4	6.6	6.6	6.9	7.7	9.4	10.3	10.7	(14)	
(15) Precipitation	PrecAvg	32.4	2.4	1.6	1.8	2.4	3.0	2.9	3.0	3.3	3.4	3.6	2.8	2.4	(15)	
	PrecMax	37.6	4.3	2.7	4.4	4.3	5.2	6.4	5.0	5.8	6.4	7.7	4.8	4.8	(16)	
	PrecMin	27.3	1.0	0.5	0.3	0.5	0.9	0.5	1.5	0.9	0.9	1.5	1.2	0.4	(17)	
	PrecStd	3.1	0.9	0.6	1.0	0.9	1.1	1.4	1.1	1.3	1.3	1.7	1.1	1.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.2	48.3	69.4	74.8	83.5	88.2	89.1	86.9	85.7	77.5	65.1	54.5	(19)	
		MCWB	43.1	43.7	58.3	57.6	67.1	73.1	73.6	71.3	70.8	66.2	55.4	51.7	(20)	
	2%	DB	41.4	42.7	56.7	66.5	77.1	82.3	84.2	82.4	81.1	72.2	57.4	44.8	(21)	
		MCWB	38.1	37.9	49.8	54.8	64.1	68.8	70.6	70.1	68.2	63.2	51.4	41.5	(22)	
	5%	DB	37.4	37.2	50.3	61.2	71.9	79.0	81.8	80.3	77.2	66.5	53.6	41.7	(23)	
		MCWB	35.3	34.3	44.3	49.8	60.7	66.6	69.6	68.8	67.2	60.0	48.4	39.3	(24)	
	10%	DB	35.2	34.3	44.8	55.0	66.2	74.9	79.2	77.3	73.3	61.8	49.7	38.8	(25)	
		MCWB	33.9	32.0	39.2	46.0	58.0	64.9	68.1	67.3	65.5	56.2	45.9	36.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.2	45.1	58.8	61.3	69.7	74.2	76.1	74.4	73.5	69.5	57.4	51.0	(27)	
		MCDB	45.3	47.9	68.1	69.2	79.6	84.3	85.2	81.9	80.1	74.2	63.2	53.6	(28)	
	2%	WB	38.6	38.3	51.2	56.0	66.4	71.1	73.6	72.5	71.1	65.1	52.3	42.6	(29)	
		MCDB	41.1	41.5	55.1	65.1	74.3	80.3	80.9	79.5	77.3	69.9	56.8	43.9	(30)	
	5%	WB	35.7	34.8	44.1	51.1	62.8	68.5	71.8	71.0	69.2	60.3	49.1	39.3	(31)	
		MCDB	37.4	37.4	49.4	59.6	69.6	76.1	78.5	77.3	75.0	66.1	53.2	41.2	(32)	
	10%	WB	33.5	32.2	39.7	46.9	59.0	66.3	70.1	69.3	66.9	56.7	45.7	36.8	(33)	
		MCDB	35.4	33.8	44.2	54.7	65.2	73.2	76.8	74.8	72.5	61.3	49.0	38.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	12.0	14.9	16.2	18.6	17.5	16.5	15.4	15.9	12.9	10.4	8.3	(35)	
		MCDBR	12.1	14.5	21.7	26.9	25.2	21.9	19.8	18.5	20.0	20.9	17.3	12.7	(36)	
	5% WB	MCWBR	10.7	12.1	15.3	16.2	14.1	11.4	10.0	9.6	10.7	13.1	12.5	11.5	(37)	
		MCWBR	11.3	14.2	20.7	25.4	22.9	18.7	16.4	15.1	17.0	18.6	16.0	11.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.291	0.293	0.318	0.351	0.382	0.392	0.389	0.383	0.371	0.340	0.304	0.280	(40)	
		taud	2.375	2.345	2.382	2.357	2.348	2.351	2.387	2.404	2.420	2.484	2.518	2.464	(41)	
	Ebn at Noon		260	283	290	288	281	278	277	275	269	263	256	254	(42)	
		Edh at Noon	23	29	33	37	38	38	37	35	31	26	21	20	(43)	
(44) All-Sky Solar Radiation	RadAvg		371	665	1125	1462	1786	1942	1954	1661	1274	744	399	273	(44)	
	RadStd		40	68	118	148	154	139	76	103	109	87	57	28	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page